HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Mathematics

F18CD

Multivariable Calculus and Analysis A

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2023

Duration: 2 hours

Answer All Questions

Attached: No additional attachments
Supplied: No additional material supplied
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

1. (a) Write down the definition of a complete ordered field. You do not need to state all the axioms for an ordered field.

[3 marks]

(b) Prove that for every $x, y \in \mathbb{R}$, with $0 < x < y$, there exists an $n \in \mathbb{N}$ such that $y < nx$.

[7 marks]

(c) Let A be the set

$$A = \left\{ 5 - \frac{2}{n} : n \in \mathbb{N} \right\} \subset \mathbb{R}.$$

Prove that the supremum of A is 5. Is 5 also the maximum of A?

[5 marks]

(d) Let $B \subset \mathbb{Z} \subset \mathbb{R}$ such that B is non-empty and bounded above. Does the maximum of B always exist? If so, provide a proof. Otherwise provide a counter-example.

[5 marks]

2. (a) Write down the definition of a convergent sequence (a_n) to converge to $a \in \mathbb{R}$.

[3 marks]

(b) Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$. Prove that

$$\lim(a_n + b_n) = a + b.$$

[7 marks]

(c) Consider the sequences (a_n) and (b_n) given by

$$a_n = \frac{n}{n+2}$$

and

$$b_n = \frac{n^2 - 3n}{2n^2 - 1}.$$

Using the algebra of limits, provide guesses for what (a_n) and (b_n) converge to and then explicitly prove that both sequences indeed converge to your guesses directly from the definition of a convergent sequences.

[7 marks]

(d) Let (a_n) and (b_n) be sequences such that (a_n) is convergent and (b_n) is divergent. Does it follow that $(a_n b_n)$ is divergent? If so, provide a proof. Otherwise, provide a counterexample.

[3 marks]

3. (a) Write down the definition of a series $\sum_{k=1}^{\infty} a_k$ to converge to a limit $L \in \mathbb{R}$.

[3 marks]

(b) Prove that if a series converges absolutely, then it converges.

[7 marks]

(c) Consider the series

$$\sum_{k=1}^{\infty} \frac{\cos(k)}{k^4}.$$

Using part (b), or otherwise, prove that the series is convergent. Make sure to explicitly state any of the series tests used to arrive at your result.

[4 marks]

(d) Prove that the series

$$\sum_{k=1}^{\infty} (-1)^k \left(\sqrt{k+1} - \sqrt{k} \right)$$

is conditionally convergent.

[6 marks]

4. (a) Let $f : D \to \mathbb{R}$ where $D \subset \mathbb{R}$. Write down the definition of the function f to be continuous at a point $c \in D$

[3 marks]

(b) Let $f : D \to \mathbb{R}$ with $D \subset \mathbb{R}$. Prove that if f is differentiable at $c \in D$, then f is continuous at $c \in D$.

[6 marks]

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be a real function given by

$$f(x) = \begin{cases} \dfrac{3x^2 - 12}{x - 2}, & \text{if } x \neq 2 \\ 12, & \text{if } x = 2 \end{cases}.$$

Prove directly from the definition of continuity that $f(x)$ is continuous at $x = 2$.

[7 marks]

(d) Suppose that $f : [a, b] \to \mathbb{R}$ is a real function such that for every $\alpha \in \mathbb{R}$, with $f(a) < \alpha < f(b)$, there exists a $c \in (a, b)$ with $f(c) = \alpha$. Does it follow that f is continuous on $[a, b]$? If so, provide a proof. Otherwise, provide a counterexample.

[4 marks]

5. (a) By considering the power series

$$\sum_{k=0}^{\infty} a_k x^k,$$

define what the radius of convergence is for such a power series.

[3 marks]

(b) Prove that the power series

$$\sum_{k=1}^{\infty} \frac{x^k}{k}$$

has radius of convergence 1.

[7 marks]

(c) Let $f : (-1, 1) \to \mathbb{R}$ given by $f(x) = \ln(1 + x)$. Given that

$$\ln(1 + x) = a_0 + a_1 x + a_2 x^2 + \ldots$$

converges for each $x \in (-1, 1)$ and some $a_k \in \mathbb{R}$, find the values of $a_k \in \mathbb{R}$ explicitly. Does this power series also converge for when $x = 1$ or $x = -1$? In each case, explain your reasoning.

[6 marks]

(d) Let $f : \mathbb{R} \to \mathbb{R}$ be a real function such that f is infinitely differentiable on its entire domain. Does there always exist a convergent power series such that

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

for all $x \in \mathbb{R}$? If so, provide a proof. Otherwise, provide a counterexample.

[4 marks]